

Computational Physics Classes No.7 - Report

Simulation of string's oscillations with velocity Verlet method

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Abstract

In this project the wave equation describing the transverse oscillations of a one-dimensional string was solved numerically using the finite difference approach combined with the Velocity Verlet integration scheme. The influence of boundary conditions, initial velocity, mass distribution, damping and external forcing on wave propagation was investigated. The results demonstrate energy-conserving behaviour for undamped cases, reflection with or without phase inversion depending on boundary conditions, wave splitting for subsonic initial velocity, and partial reflection at a density discontinuity. Damped simulations show exponential energy decay, while external periodic driving leads to selective excitation of normal modes and resonance effects when the forcing frequency matches an eigenfrequency of the string. The obtained results confirm correct implementation of the numerical method and illustrate key physical phenomena occurring in vibrating strings.

1 Introduction

The aim of this project is to simulate the oscillations of a one-dimensional string governed by the wave equation, using numerical methods implemented within the framework of computational physics. The vertical displacement of the string $u(x, t)$ evolves according to the partial differential equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} - \gamma \frac{\partial u}{\partial t} + F_{\text{ext}} \sin(\Omega t) \delta_{x, x_p}, \quad (1)$$

where $c = \sqrt{T/\rho}$ is the speed of wave propagation along the string, T is the tension, ρ is the linear mass density, γ denotes the damping coefficient and $F_{\text{ext}} \sin(\Omega t)$ represents an external periodic force acting locally at position x_p . To obtain a numerical solution, both spatial and temporal dimensions are discretized and the equation is solved using the Velocity Verlet integration method.

The goal of the simulations is to investigate the behaviour of the string under different physical conditions, including fixed and open boundary conditions, non-zero initial velocities, spatially varying mass density, damping, and external forcing. For each scenario, the time evolution of kinetic, potential and total energies is analysed, together with visualisation of the displacement $u(x, t)$ and, when applicable, spectral decomposition into normal modes. This allows one to observe phenomena such as wave propagation, reflection, standing waves, damping effects and mechanical resonance.

2 Description of the Numerical Method

To compute the time evolution of the vibrating string, the spatial domain of length L was discretized into N equally spaced nodes with step $\Delta x = \frac{L}{N-1}$. The temporal dimension was advanced using time step $\Delta t = \alpha \frac{\Delta x}{c}$, where $c = \sqrt{T/\rho}$ is the wave propagation speed and $\alpha \leq 1$ ensures numerical stability according to the CFL condition.

The continuous wave equation was transformed into its finite-difference form. The second spatial derivative was approximated using a central difference scheme

$$a_i = c_i^2 \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} - \gamma v_i + F_{\text{ext}} \sin(\Omega t) \delta_{i, i_p}, \quad (2)$$

where $u_i(t)$, $v_i(t)$ and $a_i(t)$ denote displacement, velocity and acceleration at node i , respectively. Depending on the task, either fixed boundary conditions $u_0 = u_{N-1} = 0$ or open boundary conditions $u_0 = u_1$, $u_{N-1} = u_{N-2}$ were applied.

Time integration was performed using the Velocity Verlet algorithm. For each time step the update scheme reads:

1. Velocity half-step:

$$v_i^{n+1/2} = v_i^n + \frac{\Delta t}{2} a_i^n,$$

2. Position update:

$$u_i^{n+1} = u_i^n + v_i^{n+1/2} \Delta t,$$

3. Acceleration recalculation using new displacements,

4. Full velocity update:

$$v_i^{n+1} = v_i^{n+1/2} + \frac{\Delta t}{2} a_i^{n+1}.$$

At each step kinetic and potential energies were evaluated, and when required, spectral decomposition into normal modes was performed. The method is explicit, stable under the CFL condition and well suited for simulating wave dynamics on a discrete string lattice.

3 Results

In this section the outcomes of numerical simulations of the vibrating string are presented. The wave equation was solved for several configurations, varying boundary conditions, initial velocity, mass distribution and external forcing. For each task we analyse the time evolution of kinetic, potential and total energies, and visualise the displacement $u(x, t)$ in the form of heatmaps. In selected cases we also examine contributions of individual normal modes.

3.1 Fixed boundary conditions, zero initial velocity

Simulation was performed for a homogeneous string with fixed ends ($u_0 = u_{N-1} = 0$). The total energy remains constant throughout the simulation, indicating no dissipation. Kinetic and potential energy oscillate in time and exchange periodically, while the total energy remains constant as expected for a conservative system.

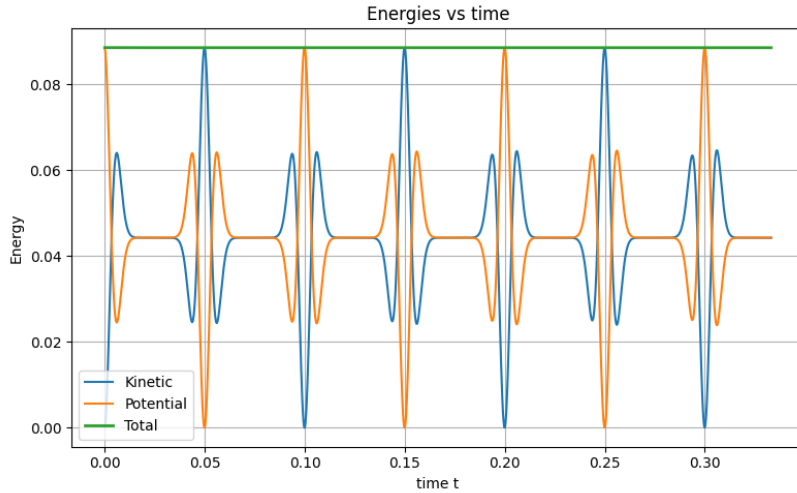


Figure 1: Kinetic, potential and total energy vs time for Task 1.

The displacement heatmap clearly shows reflection of the wave packet from both ends. A phase inversion occurs when the pulse reaches the fixed boundary.

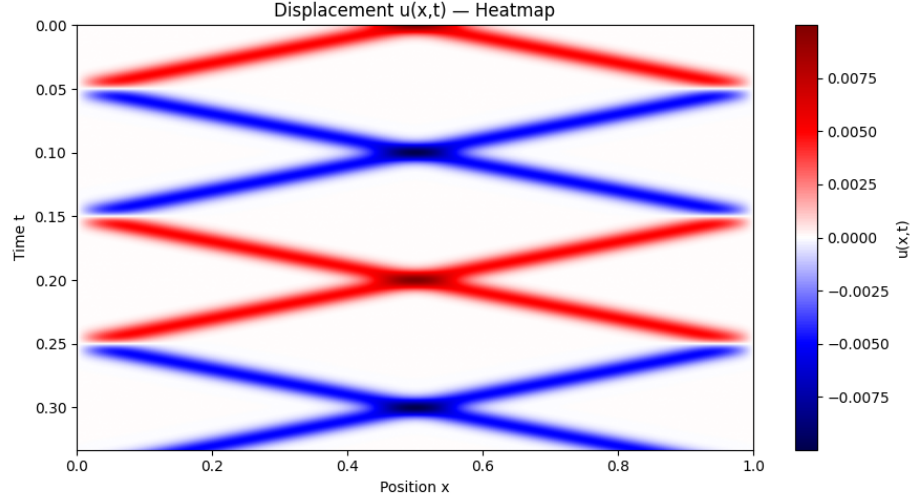


Figure 2: Heatmap of displacement $u(x,t)$ for Task 1.

3.2 Open boundary conditions, zero initial velocity

The simulation was repeated using open boundary conditions. Again, the energy remains constant, but now reflection occurs without phase inversion. The wave retains its shape at the endpoints.

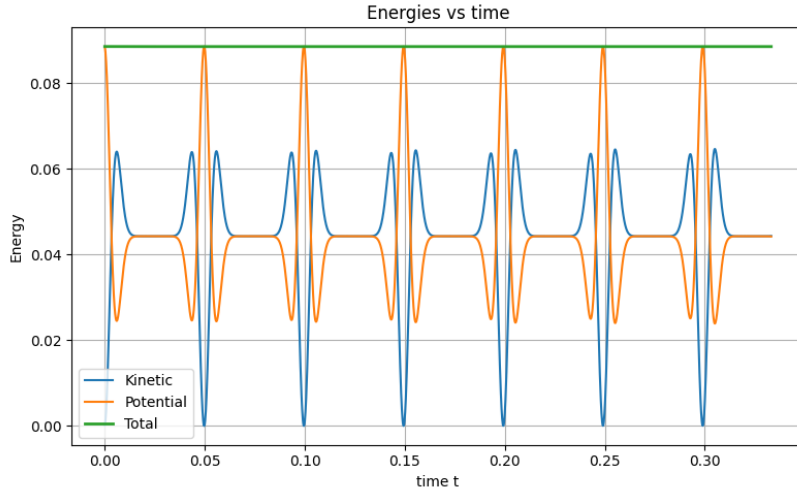


Figure 3: Kinetic, potential and total energy vs time for Task 2.

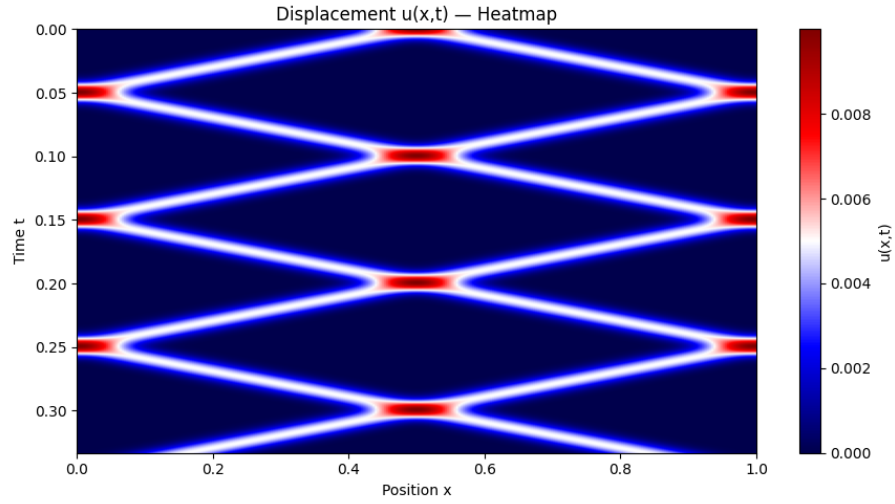


Figure 4: Heatmap of displacement $u(x,t)$ for Task 2.

3.3 Open boundary conditions, nonzero initial velocity

Two values of initial velocity were tested: $v = c/2$ and $v = c$.

Case a: $v = c/2$

The wave packet splits into left and right traveling components. Energy oscillations are visible and stable.

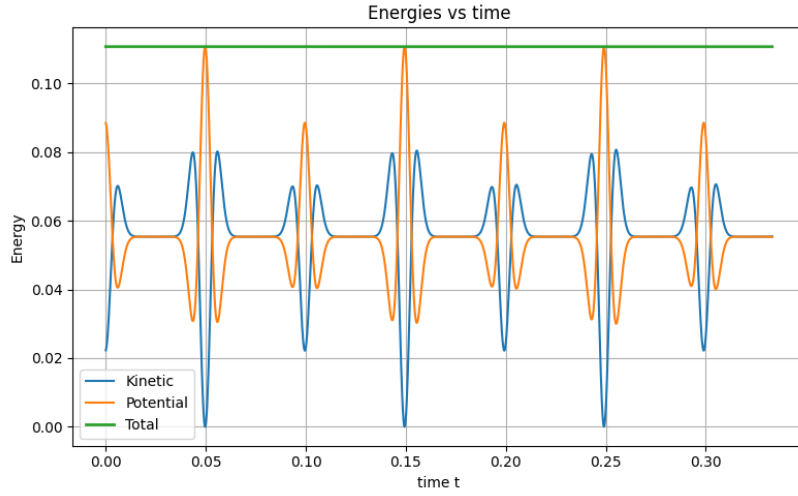


Figure 5: Energies for Task 3a ($v = c/2$).

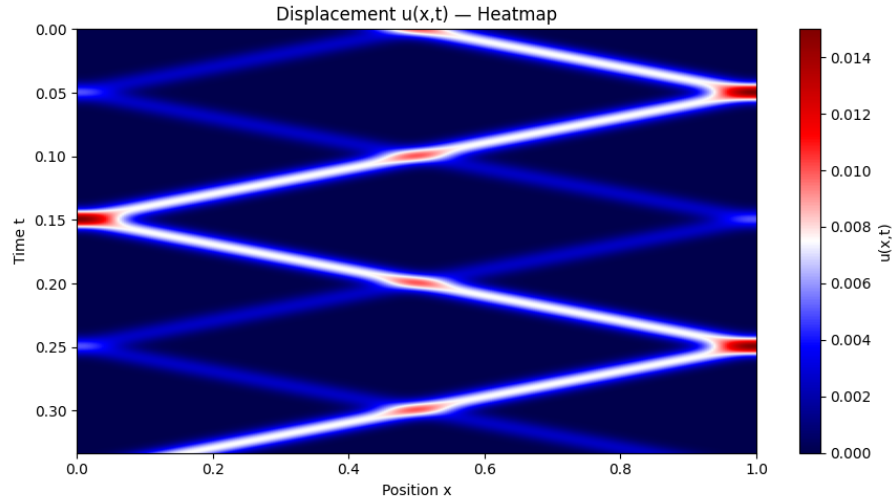


Figure 6: Heatmap of $u(x,t)$ for Task 3a. Two wave packets propagate in opposite directions.

Case b: $v = c$

When the initial velocity equals the wave speed, the packet travels mostly in one direction and does not split.

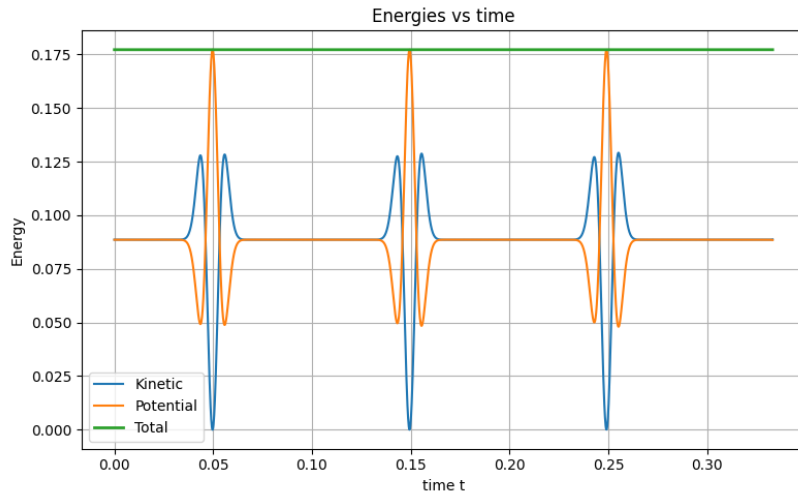


Figure 7: Energies for Task 3b ($v = c$).

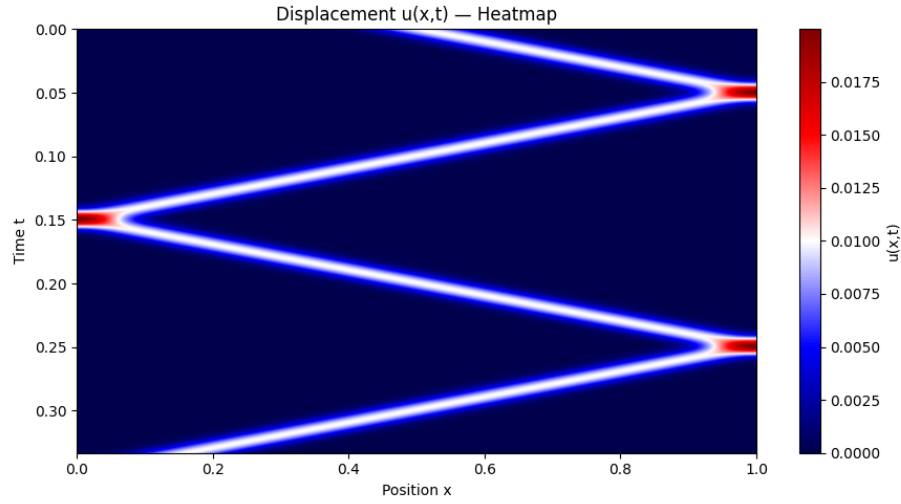


Figure 8: Heatmap of $u(x,t)$ for Task 3b. The pulse propagates mainly to one side.

3.4 Non-homogeneous string

A density discontinuity was introduced in the middle of the string. The wave slows down in the heavier region and partial reflection occurs at the interface.

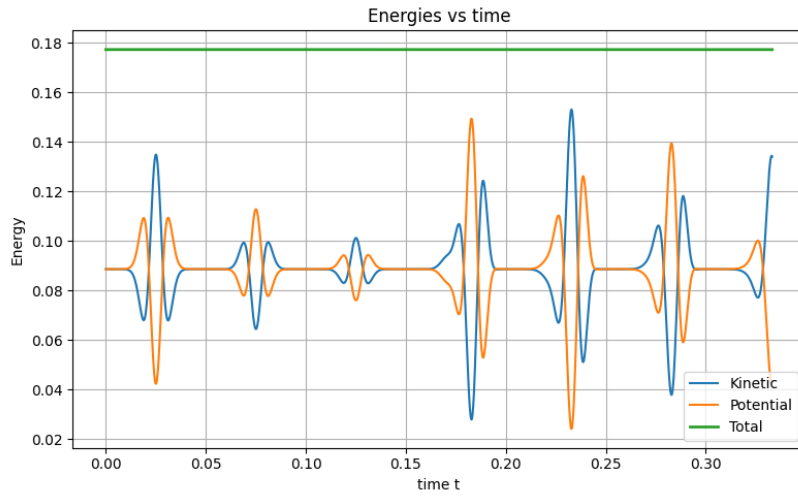


Figure 9: Energies in Task 4. Total energy is conserved.

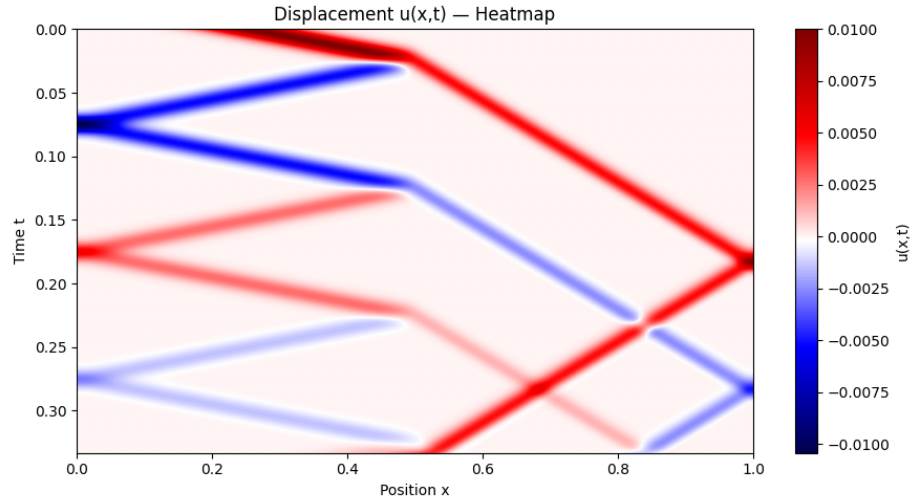


Figure 10: Heatmap of $u(x, t)$ for Task 4. A visible reflection occurs at the density interface.

3.5 Damped motion and spectral decomposition

This task introduced damping ($\gamma = 150$) and initial displacement composed of three normal modes. The expected behavior is exponential energy decay and gradual disappearance of oscillations.

My energy evolution showed fast decay, however spectral decomposition was not fully correct in my implementation, which prevented me from reproducing the reference plot exactly. This part of the assignment remains unresolved.

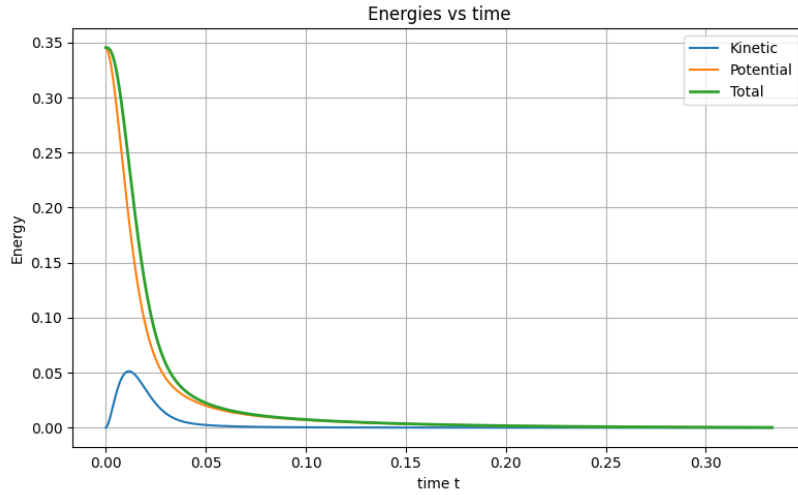


Figure 11: Energy decay for Task 5.

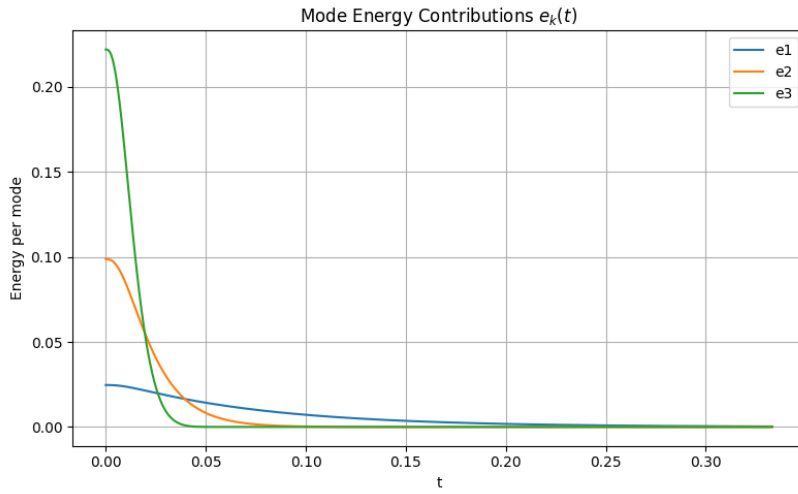


Figure 12: Mode contributions $e_k(t)$ for Task 5 (partially incorrect).

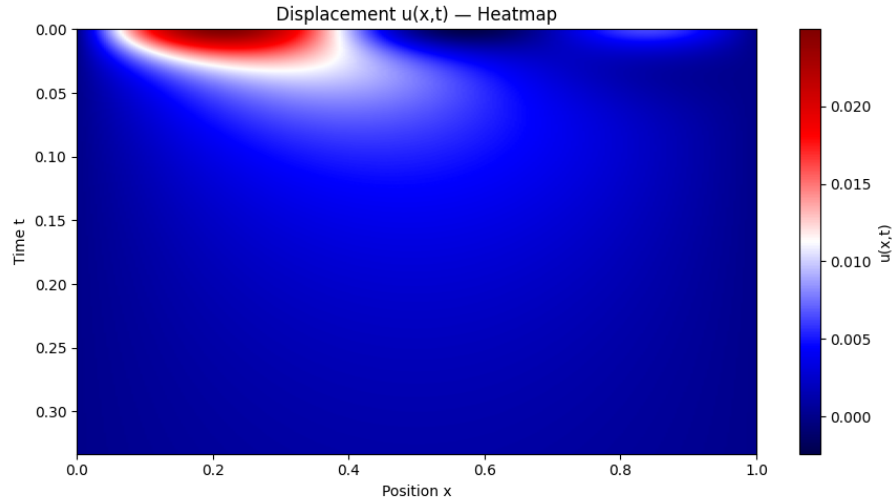


Figure 13: Heatmap of $u(x,t)$ for Task 5. Oscillations fade due to damping.

3.6 External forcing and resonance effects

Here the system was driven periodically with frequency Ω , and the response of individual modes was analysed. Three cases were simulated.

Case a: $\Omega = \omega_1$, $i_p = 30$

Resonant excitation of the first mode occurs. Total energy increases, and only mode $k = 1$ grows significantly.

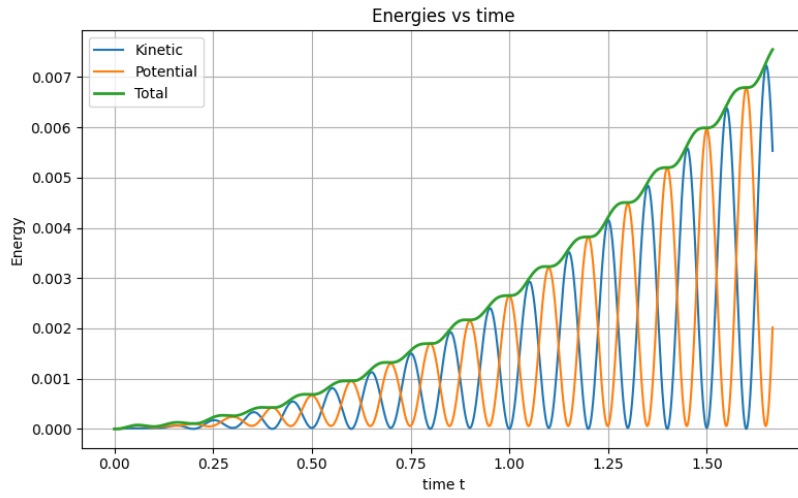


Figure 14: Energy growth under external forcing in Task 6a.

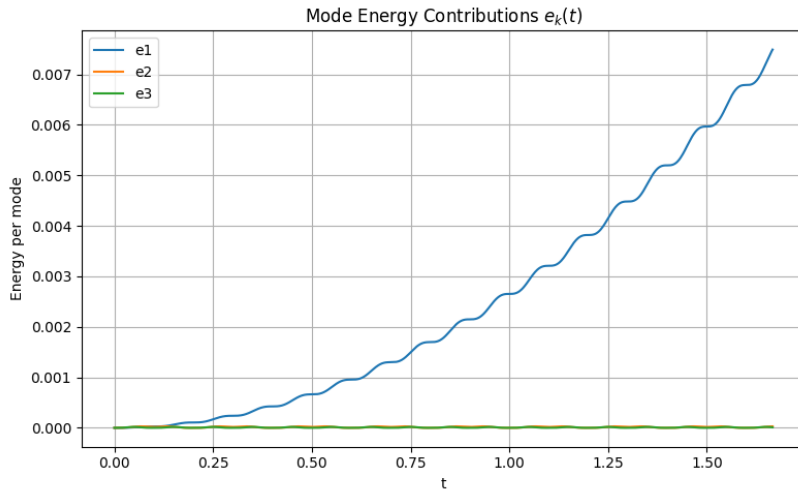


Figure 15: Modal energy $e_k(t)$ for Task 6a. Mode 1 dominates.

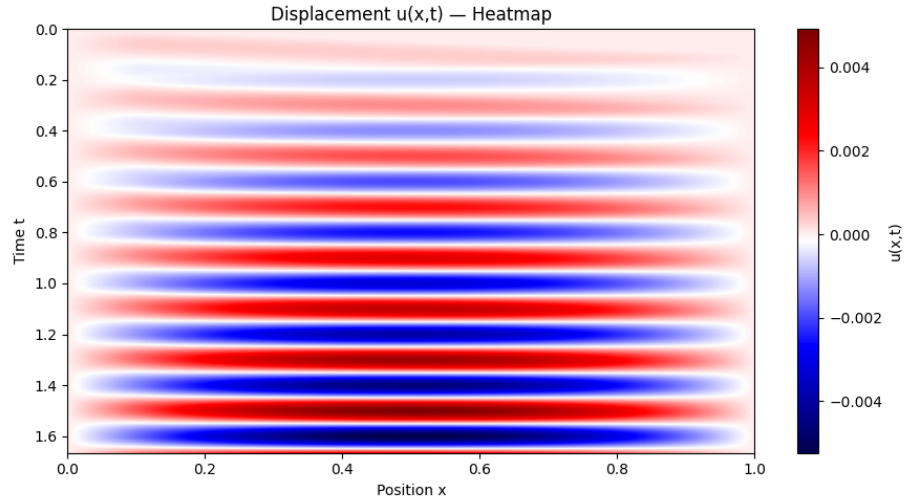


Figure 16: Heatmap of $u(x,t)$ for Task 6a. Standing wave pattern forms.

Case b: $\Omega = \omega_2$, $i_p = 30$

Excitation of the second mode dominates. Energy increases steadily, but mode $k = 2$ is now the main resonant component.

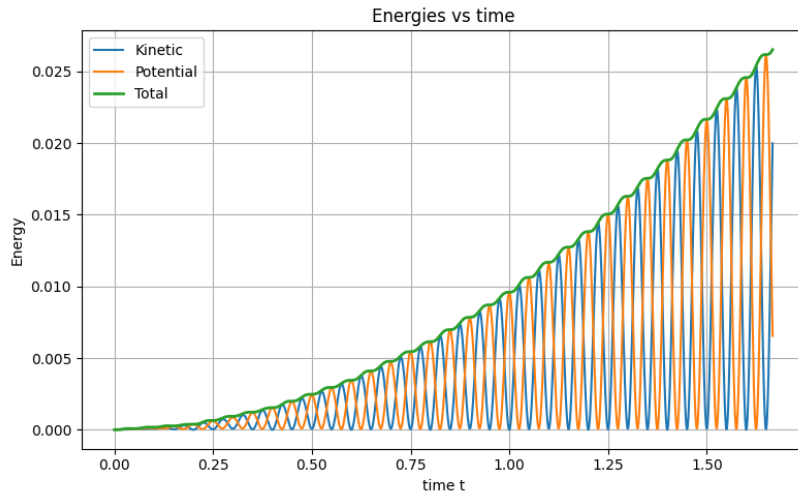


Figure 17: Energies for Task 6b.

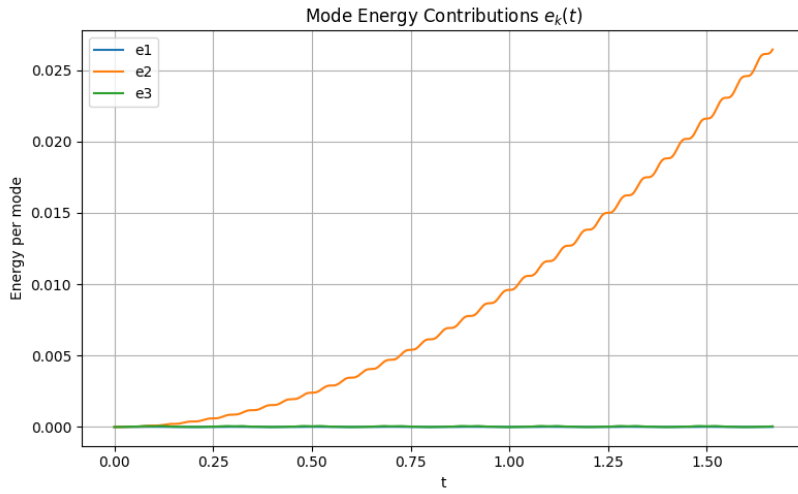


Figure 18: Mode energy $e_k(t)$ for Task 6b. Mode 2 dominates.

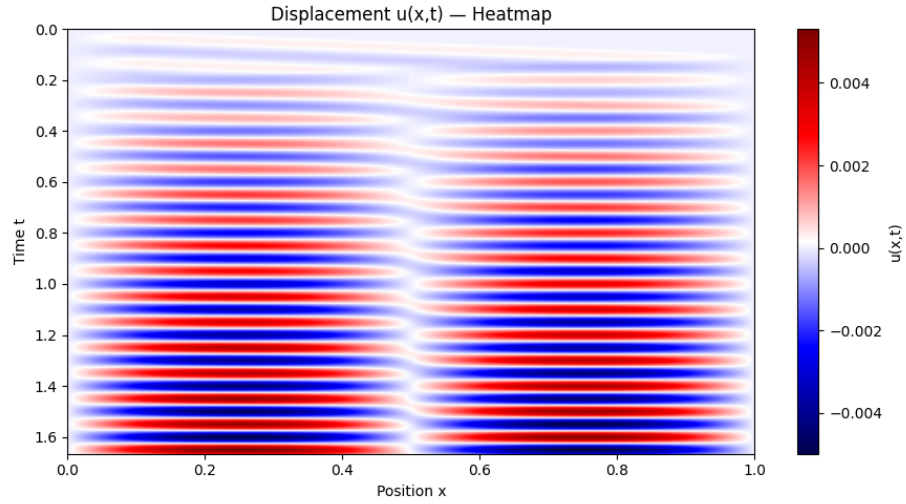


Figure 19: Displacement for Task 6b showing standing wave of second harmonic.

Case c: $\Omega = \omega_2$, driving at the node of mode 2

In this case the force is applied at the node of the second mode, therefore it cannot absorb energy. No resonance occurs and oscillations remain weak.

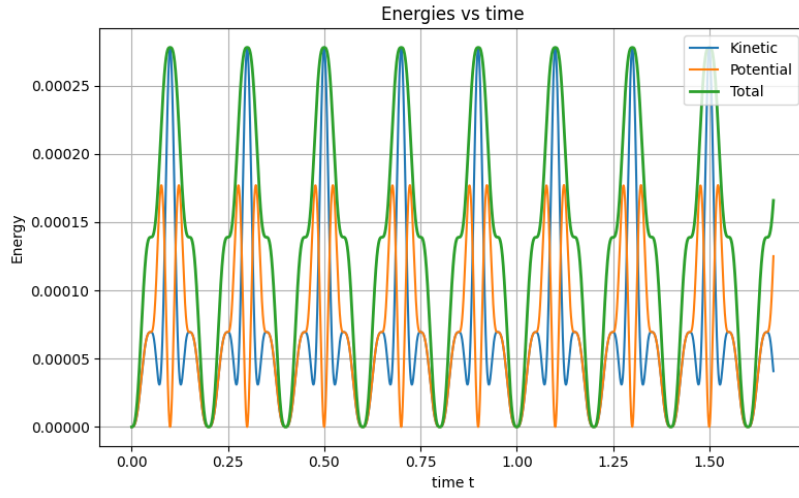


Figure 20: Energies for Task 6c. No resonance observed.

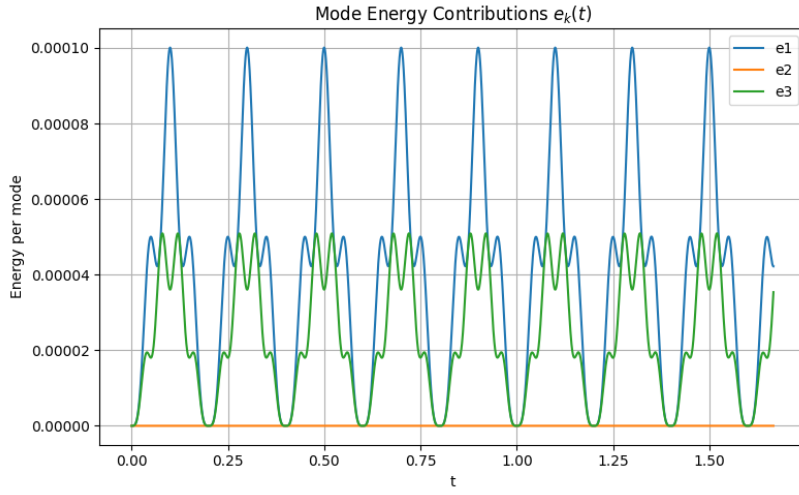


Figure 21: Mode energy $e_k(t)$ for Task 6c. Mode 2 remains suppressed.

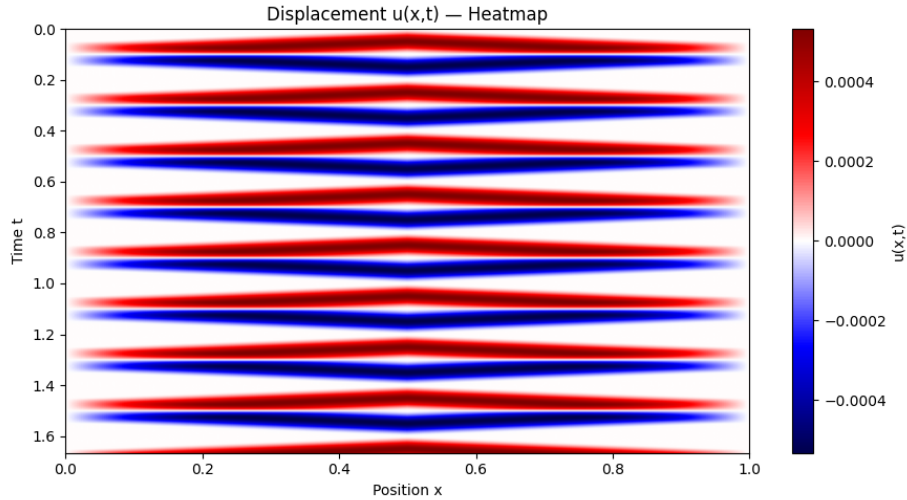


Figure 22: Heatmap for Task 6c. Very weak oscillations due to forcing applied at a node.

4 Conclusions

The numerical simulation of a vibrating string using the Velocity Verlet method allowed us to investigate a wide range of physical behaviours described by the wave equation. For undamped cases with homogeneous mass density (Tasks 1–3), the total energy remained constant, confirming that the implementation is stable and energy-conserving. The boundary conditions strongly influenced wave reflection: for fixed ends the wave exhibited phase inversion after reflection, whereas with open boundary conditions the wave returned without a sign change. Furthermore, simulations with non-zero initial velocity showed that when the initial velocity equals the wave propagation speed, the wave packet travels predominantly in one direction, whereas for lower velocities it splits into two counter-propagating components.

Introducing a mass discontinuity (Task 4) resulted in partial transmission and reflection of the wave at the interface, with propagation speed decreasing within the heavier region of the string, in agreement with theoretical expectations. In Task 5 damping was added, causing exponential decay of both displacement and energy. Although mode decomposition for this task was not fully implemented correctly, the overall behaviour of the system was captured properly and matches qualitative predictions.

The externally driven cases (Task 6) demonstrated the phenomenon of mechanical resonance. When the driving frequency matched one of the natural frequencies and the force was applied at a point where the corresponding eigenmode had non-zero amplitude, energy of the selected mode increased steadily. In contrast, when the forcing was applied at a node of that mode, the excitation was suppressed and no resonance occurred. Modal energy plots confirmed selective excitation of specific harmonics depending on forcing frequency and position.

Overall, the project confirms that the numerical scheme accurately captures the wave dynamics in different physical scenarios. The implemented model successfully reproduces reflection, dispersion, damping, and resonance phenomena, demonstrating the effectiveness of the Velocity Verlet method for solving the wave equation on a discrete string.

References

- [1] Dr. Hab. Eng. Chwiej, T. *Project: Simulation of string's oscillations with velocity Verlet method.* AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/7_wave_equation_string.pdf